



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

12-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 12-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	A,B,D	B,C,D	A,B,D	B,C,D	A,B	A,B
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	A	D	D	A	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	2	9	5	2		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	A,B,C,D	A,B,C,D	A,C,D	A,B,D	A,B,C	A,B,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	D	B	A	D	B	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	4	0	8	7		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,D	B,D	A,D	D	A,B,C,D	B,C,D	B,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	C	D	B	D	C	
SEC-3	Q.	1	2	3	4	5		
	A.	7	6	5	4	2		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (B,D)

$$r \propto \frac{1}{m}$$

so m double, radius half

$$E \propto m$$

so IE will double and gap between levels will

also double.

2. Ans (A,B,D)

When water is in liquid state

$$w_g = \frac{3}{2} mgh = k = \left[\frac{1}{2} mv^2 + \frac{1}{2} \left(\frac{2}{3} mr^2 \right) \frac{v^2}{r^2} \right] + \left[\frac{1}{2} \left(\frac{m}{2} \right) v^2 \right]$$

water will have only translational KE

when water is freezed

$$w_g = k' = \frac{3}{2} mgh = \frac{1}{2} mv'^2 + \frac{1}{2} \left(\frac{2}{3} mr^2 \right) \frac{v'^2}{r^2} + \left[\frac{1}{2} \frac{m}{2} v'^2 + \frac{1}{2} \frac{2}{5} \frac{m}{2} r^2 \frac{v'^2}{r^2} \right]$$

water will have translational as well as rotational KE so $v' < v$

3. Ans (B,C,D)

$$\rho = kr^2 = \frac{\rho_0}{R^2} r^2$$

$$(A) v_{esc} = \sqrt{\frac{2GM}{R}}$$

$$M = \int_0^R \frac{\rho_0}{R^2} r^2 4\pi r^2 dr = \frac{\rho_0 4\pi}{R^2} \frac{R^5}{5} = \frac{\rho_0 4\pi R^3}{5}$$

$$\text{so } v_{esc} = \sqrt{\frac{2G\rho_0 4\pi R^3}{5R}} = \sqrt{\frac{8\pi G\rho_0 R^2}{5}}$$

$$(B) g = \frac{GM}{r^2} = \frac{G}{r^2} \int_0^r \frac{\rho_0}{R^2} r^2 4\pi r^2 dr =$$

$$\frac{G\rho_0 4\pi r^3}{5R^2}$$

$$g \propto r^3$$

(C) as $v_{esc} = \sqrt{\frac{2GM}{R}}$ so will be same for other planet also.

$$(D) \text{ Energy required} = \frac{1}{2} m v_{esc}^2 \propto m$$

$$\text{so } \frac{E_1}{E_2} = \frac{1}{2}$$

4. Ans (A,B,D)

$$V_A = \frac{Kq_A}{R} + \frac{Kq_B}{2R} = 2V,$$

$$V_B = \frac{K(q_A + q_B)}{2R} = \frac{3}{2}V$$

$$\Rightarrow \frac{q_A}{q_B} = \frac{1}{2}$$

After earthing of B, $q_A = q_B$ &

$$V'_B = \frac{K(q'_A + q'_B)}{2R} = 0$$

$$\Rightarrow \left| \frac{q'_A}{q'_B} \right| = 1 \text{ \& } V'_A - V'_B = \frac{V}{2}$$

5. Ans (B,C,D)

$$|E| = \text{nat}^{n-1}$$

6. Ans (A,B)

$$P = \frac{P_0}{2} \left[\sin\left(203\pi t + \frac{5\pi}{6}\right) + \sin\left(197\pi t + \frac{\pi}{6}\right) \right]$$

$$f_1 = \frac{203\pi}{2\pi} > 100$$

$$f_2 = \frac{197\pi}{2\pi} < 100$$

$$f_{\text{beat}} = \frac{203\pi}{2\pi} - \frac{197\pi}{2\pi} = 3$$

7. Ans (A,B)

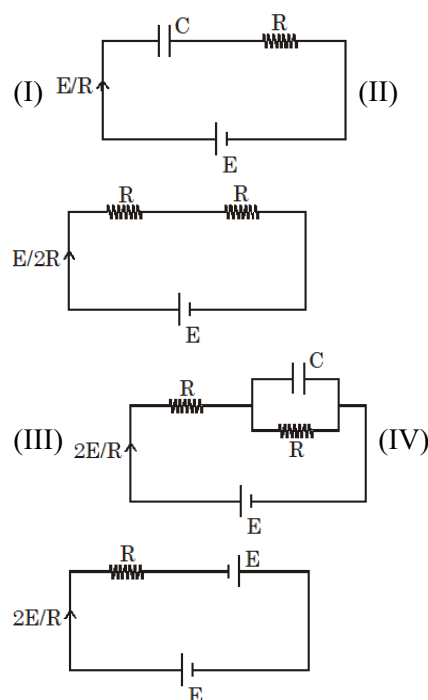
$$\delta_{\text{net}} = \delta_1 + \delta_2 + \dots + \delta_n$$

where all have same magnitude and alternate deviations are in opposite direction.

PHYSICS

SEC-2

1. Ans (B)



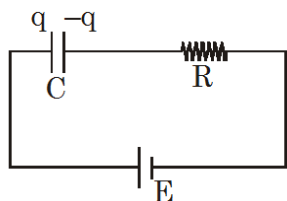
Current in III & IV is maximum

in (IV) no energy is stored as no capacitor

so ans (B) IV, i, Q

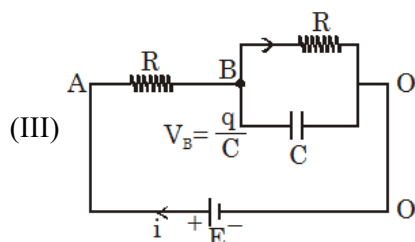
2. Ans (A)

(I) At $t = RC$



$$q = CE(1 - e^{-1}) = \text{charge drawn from battery} = 0.63 CE$$

$$(II) \text{ Charge drawn} = iRC = \frac{E}{2R}RC = 0.5CE$$



$$q = \frac{CE}{2} \left(1 - e^{-\frac{2t}{RC}}\right) \quad V_B = \frac{q}{C} \text{ \& } V_A = E$$

$$i = \frac{V_A - V_B}{R} = \frac{E}{R} - \frac{E}{2R} \left(1 - e^{-\frac{2t}{RC}}\right)$$

$$= \frac{E}{2R} + \frac{E}{2R} e^{-\frac{2t}{RC}}$$

$$\text{Charge drawn} = \int_0^{RC} i dt =$$

$$\frac{E}{2R} \int_0^{RC} \left(1 + e^{-\frac{2t}{RC}}\right) dt$$

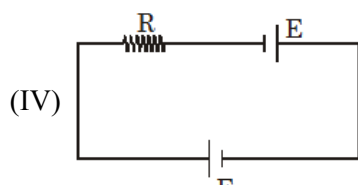
$$= \frac{E}{2R} \left[\left(t + \frac{e^{-\frac{2t}{RC}}}{(-2)}(RC)\right) \right]_0^{RC}$$

$$= \frac{E}{2R} \left[\left(RC - \frac{RC}{2}e^{-2}\right) - \left(0 - \frac{RC}{2}\right) \right]$$

$$= \frac{E}{2R} \left[\frac{3RC}{2} - \frac{RC}{2}e^{-2} \right] =$$

$$\frac{E}{2R} \frac{RC}{2} [3 - e^{-2}]$$

$$= \frac{CE}{4} (3 - e^{-2}) = 0.72 CE$$



$$\text{charge drawn} = \frac{2E}{R}RC = 2CE$$

3. Ans (D)

$$\text{in (III) } i_{t=0} = \frac{E}{R}$$

$$i_{t=\infty} = \frac{E}{2R}$$

4. Ans (D)

$$(I) F_{ABCD} \text{ due to } \rho_1 = \left(\rho_1 g \frac{h}{2}\right) h\ell;$$

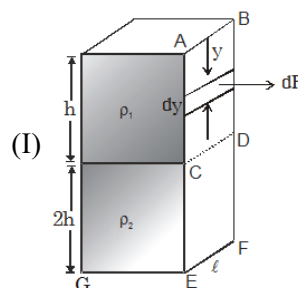
$$(II) F_{ABCD} \text{ due to } \rho_2 = 0$$

$$(III) F_{CDEF} \text{ due to } \rho_1 = \rho_1 gh(2h\ell);$$

$$(IV) F_{CDEF} \text{ due to } \rho_2 = (\rho_2 gh)(2h\ell)$$

Pressure at face CDEF due to ρ_1 is constant to effectively acts at centre

5. Ans (A)

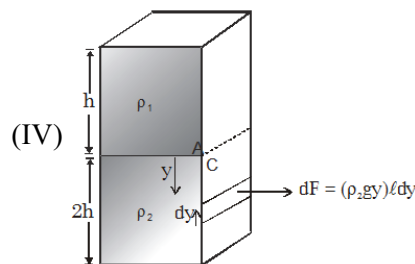


$$\int d\tau = \int_0^h (\rho_1 gy) \ell dy (3h - y)$$

$$\tau = \rho_1 g \ell \left[\frac{3hy^2}{2} - \frac{y^3}{3} \right]_0^h = \rho_1 g \ell \frac{7}{6} h^3$$

$$(II) \tau = 0$$

$$(III) \tau = (\rho_1 gh)(2h\ell)(h) = 2\rho_1 gh^3\ell$$



$$\int d\tau = \int_0^{2h} (\rho_2 g \ell y dy) (2h - y)$$

$$\Rightarrow \tau = \rho_2 g \ell \left[2h \frac{y^2}{2} - \frac{y^3}{3} \right]_0^{2h}$$

$$= \rho_2 g \ell \left[h \cdot 4h^2 - \frac{8h^3}{3} \right] =$$

$$\rho_2 g \ell \left[\frac{4h^3}{3} \right] = \frac{4\rho_2 g \ell h^3}{3}$$

$$\text{so } \frac{7}{6} \rho_1 g \ell h^3 < \frac{4}{3} \rho_2 g \ell h^3 \text{ (as } \rho_1 = \rho_2)$$

so minimum torque in (I)

6. **Ans (D)**

$$\text{Weight of liquid } \rho_2 = \rho_2(2h\ell h)g = 2\rho_2\ell h^2g$$

PHYSICS

SEC-3

1. **Ans (5)**

From COAM

$$mv_0 \frac{\ell}{6} = \left(2m \left(\frac{\ell}{3} \right)^2 + m \left(\frac{2\ell}{3} \right)^2 \right) \omega$$

$$\omega = \frac{v_0}{4\ell}$$

From COM

$$mv_0 = 3mv_{cm}$$

$$v_{cm} = \frac{v_0}{3}$$

Loss in KE

$$\begin{aligned} &= \frac{1}{2} mv_0^2 - \left\{ \frac{1}{2} (3m)v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2 \right\} \\ &= \frac{1}{2} mv_0^2 - \\ &\quad \left\{ \frac{1}{2} (3m) \left(\frac{v_0}{3} \right)^2 + \frac{1}{2} \left(\frac{2m\ell^2}{3} \right) \left(\frac{v_0}{4\ell} \right)^2 \right\} \\ &= \frac{15}{48} mv_0^2 \end{aligned}$$

3. **Ans (9)**

$$\text{Resistance of bulb, } R_{bulb} = \frac{V_{bulb}^2}{P_{bulb}} = 0.5\Omega$$

In order to the bulb glows at full intensity,

current will pass through the bulb =

$$\frac{4.5W}{1.5V} = 3A$$

equivalent resistance of circuit, $R_{eq} =$

$$\frac{5}{3} + \frac{0.5 \times 1}{0.5 + 1} = \frac{5}{3} + \frac{0.5}{1.5} = \frac{5}{3} + \frac{1}{3} = 2\Omega$$

$$\therefore \text{Current supplied by battery } \frac{E}{R_{eq}} = \frac{E}{2}$$

$$\text{Current in bulb} = I \times \frac{1}{(1 + 0.5)} \Rightarrow 3A =$$

$$\frac{I}{1.5} = \frac{E}{2 \times 1.5} \Rightarrow E = 9 \text{ volt.}$$

4. **Ans (5)**

Limiting friction between A & B = 90 N

Limiting friction between B & C = 80 N

Limiting friction between C & ground = 60 N

Since limiting friction is least between C and ground, slipping will occur at first between C and ground. This will occur when $F = 60 \text{ N}$.

5. **Ans (2)**

$$T = 2\pi \sqrt{\frac{I}{C}} \Rightarrow$$

$$\frac{T'}{T} = \sqrt{\frac{I'}{I}} = \sqrt{\frac{\left(\pi \frac{4r^2}{2} \frac{t}{4} \right) 4r^2}{\frac{(\rho \pi r^2 t) r^2}{2}}}$$

$$T' = 2T$$

CHEMISTRY

SEC-1

3. **Ans (A,B,C,D)**

Borax on heating produces H_2O (neutral oxide) + $NaBO_2 + B_2O_3$.

4. **Ans (A,C,D)**

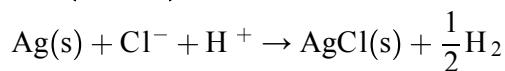
Decomposition reactions are endothermic in nature

$$\Delta_r S = \frac{\Delta_r H}{T} \text{ at equilibrium.}$$

$$\Delta_r G = \underbrace{\Delta_r H}_{+ve} - T \underbrace{\Delta_r S}_{-ve} \Rightarrow$$

(-ve) at high temperature.

5. **Ans (A,B,D)**



$$E = 0 - (0.18) - 0.06 \log \frac{1}{[H^+][Cl^-]}$$

$$= -0.18 - 0.06 \times 6 = -0.54V$$

$\therefore$ Cell will be represented as : $Pt | H_2 | HCl |$

$AgCl | Ag$ & Cell potential = 0.54 V

$$E_{Cl^-|AgCl|Ag}^\circ = E_{Ag^+|Ag}^\circ + 0.06 \log K_{sp} (AgCl)$$

$$0.18 = E_{Ag^+|Ag}^\circ - 0.06 \times 10$$

$$\Rightarrow E_{Ag^+|Ag}^\circ = 0.78 \text{ volt}$$

CHEMISTRY

SEC-2

1. **Ans (D)**

0.1 M HCOOK

$$\Delta T_f = iK_f m \simeq 2 \times 0.1 \times K_f = 0.2 K_f$$

$$\pi \simeq 0.2 RT$$

$$pOH = \frac{1}{2}[pK_w - pK_a - \log C] = \frac{1}{2}[14 - 4 + 1] = 5.5$$

$$pH = 8.5$$

Electrolysis products :

CO₂, H₂ at anode

: H₂ at cathode

2. **Ans (B)**

0.1 m Ca(NO₃)₂

$$\Delta T_b = iK_b m = 0.3 K_b$$

$$\pi \simeq 0.3 RT$$

Electrolysis products : H₂ at cathode : O₂ at anode

pH = 7 (salt of S.A + S.B)

3. **Ans (A)**

0.1 M CH₃COONa + 0.2 M CH₃COOH $\Rightarrow$ Acidic

buffer solution

$$\pi = (i_1 M_1 + i_2 M_2)RT \simeq (0.1 \times 2 + 0.2)RT$$

$$\simeq 0.4RT$$

Electrolysis products : H₂ at cathode

: CO₂, C₂H₆ at anode mainly

$$pH = pK_a + \log \frac{0.1}{0.2} = 5 - 0.3 = 4.7$$

CHEMISTRY

SEC-3

1. **Ans (5)**

B = CO₂

D = C₂H₂

2. **Ans (4)**

(iii), (iv), (v), (vii)

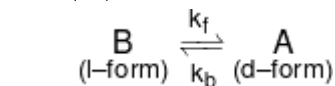
3. **Ans (0)**

x = 3 (K, Rb, Cs)

y = 2 (Mg, Be)

u = 5 (Concentrated solution of all alkali metals in liquid ammonia is blue coloured solution)

5. **Ans (7)**



$$t=0 \quad a \quad 0$$

$$t \quad a-x \quad x$$

$$t = t_{eq} \quad (a-x_{eq}) \quad x_{eq}$$

$$\text{at equilibrium} \quad \frac{dx}{dt} = k_f(a-x_{eq}) - k_b x_{eq} = 0$$

$$\Rightarrow K_{eq} = \frac{k_f}{k_b} = \frac{x_{eq}}{a-x_{eq}}$$

$$\text{Where, } x_{eq} = \frac{a}{2} \Rightarrow k_f = k_b$$

$$(k_f + k_b) = \frac{2.303}{t} \log \left(\frac{x_{eq}}{x_{eq} - x} \right)$$

$$\Rightarrow 2k_f = \frac{2.303}{230.3} \log \frac{a/2}{a/2 - a/10} = \frac{1}{100} \log \frac{0.5}{0.4}$$

$$= \frac{1}{100} (0.7 - 0.6) = 10^{-3} s^{-1}$$

$$\therefore k_f = 5 \times 10^{-4} s^{-1} = k_b$$

$$\therefore k_{eq} = 1 \quad \Delta_r G^\circ = 0, \Delta_r G = -ve$$

As A & B are enantiomers

$$\Rightarrow (S_m^\circ)_A = (S_m^\circ)_B$$

$$\text{and } \Delta_r H^\circ = 0, \Delta_r H = 0$$

MATHEMATICS

SEC-1

1. **Ans (B,D)**

$$\vec{a} \times \vec{b} = x\vec{a} + y\vec{b} + z\vec{c}. \text{ Let } \vec{a} = \lambda\vec{b} \times \vec{c}. \lambda = \frac{1}{2}$$

$$\text{So, } \frac{\vec{b} \times \vec{c}}{2} \times \vec{b} = \frac{x}{2} \vec{b} \times \vec{c} + y\vec{b} + z\vec{c}$$

$$\Rightarrow -\frac{\vec{b}}{2} + \frac{\vec{c}}{6} = \frac{x}{2} (\vec{b} \times \vec{c}) + y\vec{b} + z\vec{c}$$

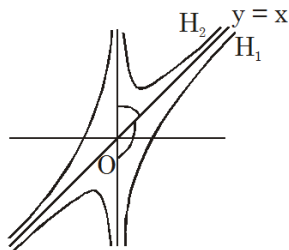
$$\Rightarrow x = 0, y = -\frac{1}{2}, z = \frac{1}{6}$$

2. **Ans (B,D)**

$$\text{Integrating we get ; } x^2 + (f(x))^2 + (f'(x))^2 = k$$

so, f(x) and f'(x) are bounded

3. Ans (A,D)



asymptote of $x^2 = xy + 1$ and $x^2 + 1 = xy$

are $x^2 - xy = 0$

and $y^2 - xy - 1 = 0$ and $x^2 - xy + 1 = 0$ are conjugate of each other

$$\Rightarrow \frac{1}{e_1^2} + \frac{1}{e_2^2} = 1 \text{ (direct result)}$$

for $x^2 + 1 = xy$ if we rotate the conic by $\frac{\pi}{8}$ it

$$\text{will be of the form } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\text{with asymptotes } y = \pm \frac{b}{a}x \Rightarrow \frac{b}{a} = \tan \frac{\pi}{8}$$

$$\Rightarrow e_2^2 = 1 + \frac{b^2}{a^2} = \frac{1}{1 + \cos \frac{\pi}{8}}$$

$$\Rightarrow \frac{1}{e_1^2} = 1 - \frac{1}{e_2^2} = 1 - \frac{1}{\sec \frac{\pi}{8}} = 1 - \cos \frac{\pi}{8}$$

$$\Rightarrow \left. \begin{aligned} e_1^2 &= \frac{1}{1 - \cos \frac{\pi}{8}} \\ e_2^2 &= \frac{1}{1 + \cos \frac{\pi}{8}} \end{aligned} \right\} \Rightarrow e_1 > e_2$$

4. Ans (D)

$$p = \frac{2^{2 \times 2}}{2^{3 \times 3}} = \frac{1}{32}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Total number of matrices = 2^9

$(s_1 s_2 s_3 t_1 t_2 t_3)$ is odd integer if

(1) all the elements are 1 (no of matrices = 1)

(2) four zeros and five ones (no of matrices = 9)

(zeros must occupy all the positions of any minor)

(3) six zeroes and three ones (no of matrices = 6)

$$({}^3C_1 \cdot {}^2C_1 \cdot 1)$$

$$\Rightarrow \text{required probability} = \frac{16}{2^9} = \frac{2^4}{2^9}$$

5. Ans (A,B,C,D)

Given $AA^T = I$ and $|A| = 1$

$$(A) |A - I| = |A - AA^T| = |I - A^T|$$

$$= |(I - A^T)^T|$$

$$|A - I| = |I - A| = -|A - I|$$

$$\Rightarrow |A - I| = 0$$

$$(B) |A^2 - I| = |A^2 - AA^T|$$

$$= |A| |A - A^T| = |A - A^T|$$

since $A - A^T$ is skew symmetric

$$\Rightarrow |A - A^T| = 0 \Rightarrow \text{hence is not invertible}$$

$$(C) |A - A^T| = 0 \Rightarrow (A - A^T)^2 \text{ is not invertible}$$

$$(D) (A - (\det A)) = A - I \text{ is not invertible}$$

6. Ans (B,C,D)

Replace x by $-x$ we get ;

$$(2x^2 + 2x + 1)^n = \sum_{r=0}^{2n} a_r x^r$$

$$\text{So, } \sum_{r=0}^{2n} a_r = 5^n, \sum_{r=0}^{2n} (-1)^r a_r = 1. \text{ Also, } a_0 = 1$$

again,

$$\begin{aligned} \sum_{r=0}^{2n} a_r x^r &= ((x+1)^2 + x^2)^n = \sum^n c_k x^{2k} (x+1)^{2n-2k} \\ &= \sum \left[{}^n c_k \sum {}^{2n-2k} c_t x^{2n-t} \right] \end{aligned}$$

$\Rightarrow$ comparing coeff. of x^{2n-r} , we get

$$\sum^n c_k {}^{2n-2k} C_r = a_{2n-r} = 2^{n-r} a_r$$

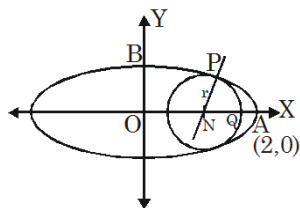
$$\text{Also, } \sum_{r=0}^{2n} (-1)^r \cdot a_r a_{2n-r} = \text{coeff. of } x^{2n}$$

$$\text{in } ((2x^2 - 2x + 1)^n (2x^2 + 2x + 1)^n)$$

$$= \text{coeff. of } x^{2n} \text{ in } (4x^4 + 1)^n = 0 \text{ if } n \text{ odd.}$$

7. Ans (B,C)

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$



$N \equiv (k, 0)$ & $Q(r+k, 0)$

Normal PN at

$P(2 \cos \theta, \sqrt{3} \sin \theta)$; $\theta \neq \frac{n\pi}{2}$ is

$$2x \sec \theta - \sqrt{3}y \csc \theta = 1$$

$$\Rightarrow N \equiv \left(\frac{\cos \theta}{2}, 0 \right) \Rightarrow k = \frac{\cos \theta}{2} \in \left(0, \frac{1}{2} \right)$$

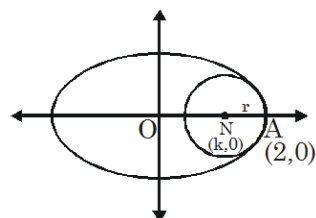
[as $k > 0$ & $\theta \neq \frac{n\pi}{2}$]

So, all such circles touches the arc of the ellipse in the Ist (as well as 4th) quadrant between (strictly) A & B.

Thus, $0 < k < \frac{1}{2}$ & $r+k < 2$ (as $Q_x < A_x$)

Now, there is another possibility when point of normalcy is A (note it cannot be B otherwise $k=0$)

In this case,



$$r+k=2$$

So, circle is

$$(x-k)^2 + y^2 = (2-k)^2$$

Intersecting it with given ellipse :

$$3x^2 + 4[(2-k)^2 - (x-k)^2] = 12$$

$$\Rightarrow x^2 - 8kx + 16k - 4 = 0$$

$$\Rightarrow x = 2, 8k - 2$$

Since, the circle lies completely inside the ellipse

So, other point of intersection

(i.e. $x = 8k - 2$) should be imaginary.

Hence, $8k - 2 > 2$ (Note $8k - 2 > -2$ as $k > 0$)

$$\Rightarrow k > \frac{1}{2} \text{ & } r+k=2$$

MATHEMATICS

SEC-2

1. Ans (C)

$x + y = 1$ represents a line

$x^2 + y^2 = 1$ represents a circle one of whose

tangents is $x + y = \sqrt{2}$ at point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$

where as, $\sqrt{x} + \sqrt{y} = 1 \Rightarrow y = 1 + x - 2\sqrt{x}$

$\Rightarrow 4x = (1 + x - y)^2$ represents a parabola.

and $\frac{1}{x} + \frac{1}{y} = 1$ represents a hyperbola.

2. Ans (C)

$x + y = 1$ represents a line

$x^2 + y^2 = 1$ represents a circle one of whose

tangents is $x + y = \sqrt{2}$ at point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$

where as, $\sqrt{x} + \sqrt{y} = 1 \Rightarrow y = 1 + x - 2\sqrt{x}$

$\Rightarrow 4x = (1 + x - y)^2$ represents a parabola.

and $\frac{1}{x} + \frac{1}{y} = 1$ represents a hyperbola.

3. Ans (D)

$x + y = 1$ represents a line

$x^2 + y^2 = 1$ represents a circle one of whose

tangents is $x + y = \sqrt{2}$ at point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$

where as, $\sqrt{x} + \sqrt{y} = 1 \Rightarrow y = 1 + x - 2\sqrt{x}$

$\Rightarrow 4x = (1 + x - y)^2$ represents a parabola.

and $\frac{1}{x} + \frac{1}{y} = 1$ represents a hyperbola.

4. Ans (B)

(P) $[x] \{x\} = 1 \Rightarrow x \in$

$$\left\{ n + \frac{1}{n} : n \in \mathbb{I}, n \geq 2 \right\}$$

(Q) $[x] \{x\} = 1 \Rightarrow x \in$

$$\left\{ n + \frac{1}{|n|} : n \in \mathbb{I}, |n| \geq 2 \right\}$$

$$(R) \left[x \left[\frac{1}{x} \right] \right] = 1 \Rightarrow x \in (-2, 0) \cup \left\{ \frac{1}{n} : n \in \mathbb{I}^+ \right\}$$

$$(S) \{x\} + \left\{ \frac{1}{x} \right\} = 1 \Rightarrow x \in \left\{ \frac{n \pm \sqrt{n^2 - 4}}{2} : n \in \mathbb{I}, |n| \geq 3 \right\}$$

5. Ans (D)

$$(P) [x] \{x\} = 1 \Rightarrow x \in$$

$$\left\{ n + \frac{1}{n} : n \in I, n \geq 2 \right\}$$

$$(Q) [x] \{x\} = 1 \Rightarrow x \in$$

$$\left\{ n + \frac{1}{|n|} : n \in I, |n| \geq 2 \right\}$$

$$(R) \left[x \left[\frac{1}{x} \right] \right] = 1 \Rightarrow x \in (-2, 0) \cup \left\{ \frac{1}{n} : n \in I^+ \right\}$$

$$(S) \{x\} + \left\{ \frac{1}{x} \right\} = 1 \Rightarrow x \in \left\{ \frac{n \pm \sqrt{n^2 - 4}}{2} : n \in I, |n| \geq 3 \right\}$$

6. Ans (C)

$$(P) [x] \{x\} = 1 \Rightarrow x \in$$

$$\left\{ n + \frac{1}{n} : n \in I, n \geq 2 \right\}$$

$$(Q) [x] \{x\} = 1 \Rightarrow x \in$$

$$\left\{ n + \frac{1}{|n|} : n \in I, |n| \geq 2 \right\}$$

$$(R) \left[x \left[\frac{1}{x} \right] \right] = 1 \Rightarrow x \in (-2, 0) \cup \left\{ \frac{1}{n} : n \in I^+ \right\}$$

$$(S) \{x\} + \left\{ \frac{1}{x} \right\} = 1 \Rightarrow x \in \left\{ \frac{n \pm \sqrt{n^2 - 4}}{2} : n \in I, |n| \geq 3 \right\}$$

MATHEMATICS

SEC-3

1. Ans (7)

at $A = \frac{\pi}{7}, B = \frac{2\pi}{7}, C = \frac{4\pi}{7}$ which are angles of a triangle.

So, $x = \tan A, y = \tan B, z = -\tan C$.

$$\frac{1}{2}(x^2 + y^2 + z^2 - x^2 y^2 z)$$

$$= \frac{1}{2} [\tan^2 A + \tan^2 B + \tan^2 C - (\tan A \tan B \tan C)^2]$$

$$= -(\tan A \tan B + \tan B \tan C + \tan C \tan A)$$

$$= -\frac{\sin A \sin B \cos C + \sin B \sin C \cos A + \sin C \sin A \cos B}{\cos A \cos B \cos C}$$

$$= -\frac{\cos(A + B + C) - \cos A \cos B \cos C}{\cos A \cos B \cos C}$$

$$= -\frac{1}{\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}} - 1 = 8 - 1 = 7.$$

2. Ans (6)

$$f'(x) = 0 \Rightarrow x = -\frac{9}{5a}, \frac{1}{a}$$

As, $a < 0$ and maxima comes first so,

$$\frac{1}{a} = -\frac{9}{5} \Rightarrow a = -\frac{5}{9} \text{ and } \alpha = 1$$

$$\text{Now, } f(1) > 0 \Rightarrow b > \frac{36}{5} \Rightarrow a + b > \frac{27}{5}$$

3. Ans (5)

$$\text{Let } t = \frac{x^3 - 3x + 1}{x^2 - x} = \frac{(x^2 - x)(x + 1) - (x - 1) - x}{x^2 - x}$$

$$= x + 1 - \frac{1}{x} - \frac{1}{x - 1}$$

So, $I =$

$$\int_{-\frac{17}{12}}^{-\frac{6}{12}} \frac{1}{t^2} dt = \left(-\frac{1}{t} \right) \Big|_{-\frac{17}{12}}^{-\frac{6}{12}} = 6 - \frac{12}{17} = \frac{90}{17}$$

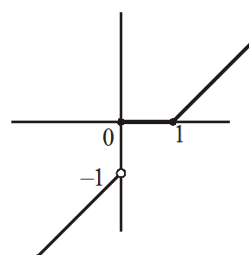
$$(\text{for } x = -2, t = -\frac{1}{6} \text{ and for } x = -3, t = -\frac{17}{12})$$

4. Ans (4)

g is symmetrical about

$$x = \frac{\pi}{2} \cdot g_{\max} = g\left(\frac{\pi}{2}\right) = 2f\left(\frac{\pi}{2}\right) = 4$$

5. Ans (2)



Not differentiable at $x = 0, 1$